

How to Write a Technical Paper?

By

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Types of rejection

DESK REJECTION (JOURNAL ADMINISTRATOR)

EDITOR REJECTION

REVIEWERS' REJECTION



Desk rejection

Journal administrator

Very fast

Within 1 to 3 days

Structural and formatting issues



Editor rejection



Fast

Within 1 to 2 weeks



Reasons

Out of the scope

Insignificant and insufficient scientific contribution

Results are limited

Not appropriately narrated



Reviewers' rejection



Slow

From 1 month to more than a year



Reasons

Unclear research problem

Insufficient scientific contribution

Unsound research design and methods

- No sufficient description of the research methods
- Non-reproducible experiments

Results do not support the claim/conclusion

- Results are not relevant to the research problem
- Results show minor improvement



How could the manuscript's structure help?



AVOIDING DESK REJECTION



AVOIDING EDITOR'S REJECTION



How could the manuscript's structure help?(Cont.)

- Journal administrators
  - Normally are not from the same area of research
    - They do not look into the scientific content
  - look into:
    - Formal structure of the article
    - Language
    - Format/Layout
      - Elsevier: Your Paper, Your Way (YPYW)



How could the manuscript's structure help (Cont.)

- Journal Editor
  - An expert in the field of research
  - Looks into
    - The scientific contribution
      - Title, Abstract, Introduction, Conclusion
    - The relevancy of the contribution to research community
  - Decides whether the manuscript has the merit to dedicate resources for reviewing.
    - Does not go through the article thoroughly



Structuring Your Article

Title	Abstract	Keywords
Main Body	Conclusion	References



Structuring Your Article (Cont.)

- Article's Main Body
  - Follows the IMRAD model
    - Both **journal administrator** and editor examine this structure
- Introduction
  - Some approaches put the **related work** in a separated section
- Methodology
- Results and Discussion
  - Some papers put **results and discussion** in two different sections



Structuring Your Article (Cont.)



The way we write an article is different from the way we read it.



Reading an article

Top-down approach



Writing article

Bottom-up approach



Informative Manuscript Titles

- First thing editors examine
- Titles should:
  - Attract readers attention
  - Informative but concise
  - Identifies the main issue
  - Technical jargons-free
- Normally has three elements
  - What to propose (solution)
  - Why (problem)
  - How (method)

7394 IEEE INTERNET OF THINGS JOURNAL, VOL. 8, NO. 9, MAY 1, 2021

Automatic Permission Optimization Framework for Privacy Enhancement of Mobile Applications

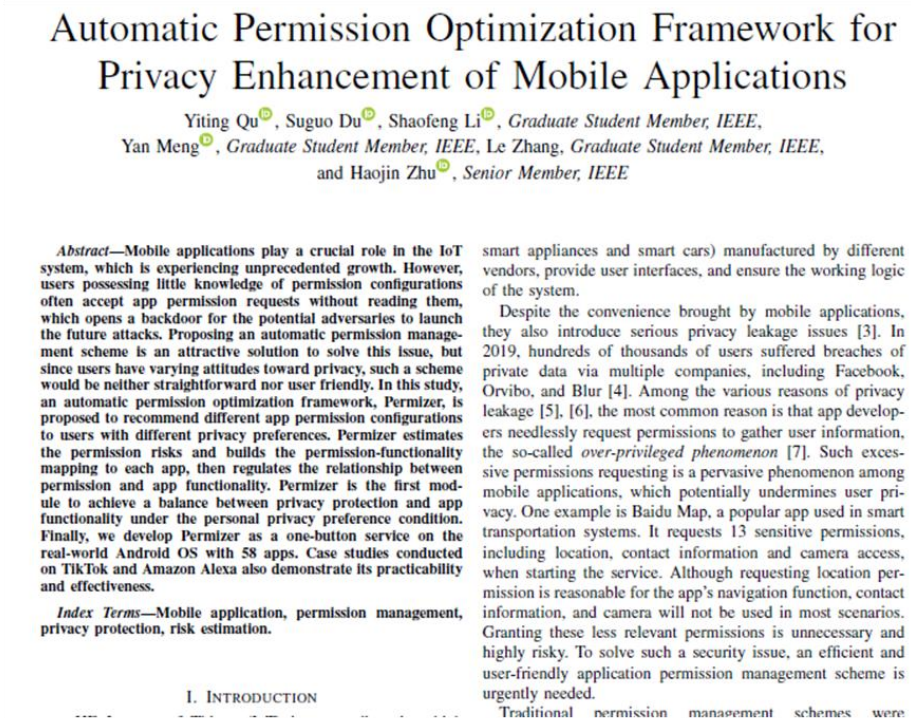
Yiting Qu, Suguo Du, Shaofeng Li, *Graduate Student Member, IEEE*, Yan Meng, *Graduate Student Member, IEEE*, Le Zhang, *Graduate Student Member, IEEE*, and Haojin Zhu, *Senior Member, IEEE*

Fast Consensus-Based Time Synchronization Protocol Using Virtual Topology for Wireless Sensor Networks



Keywords

- Are The labels of the manuscript
- Used by indexing and abstracting services
- Should be specific
- Abbreviation-free (Except the well known as DNA …etc.)



Automatic Permission Optimization Framework for Privacy Enhancement of Mobile Applications

Yiting Qu, Suguo Du, Shaofeng Li, *Graduate Student Member, IEEE*, Yan Meng, *Graduate Student Member, IEEE*, Le Zhang, *Graduate Student Member, IEEE*, and Haojin Zhu, *Senior Member, IEEE*

***Abstract*—Mobile applications play a crucial role in the IoT system, which is experiencing unprecedented growth. However, users possessing little knowledge of permission configurations often accept app permission requests without reading them, which opens a backdoor for the potential adversaries to launch the future attacks. Proposing an automatic permission management scheme is an attractive solution to solve this issue, but since users have varying attitudes toward privacy, such a scheme would be neither straightforward nor user friendly. In this study, an automatic permission optimization framework, Permizer, is proposed to recommend different app permission configurations to users with different privacy preferences. Permizer estimates the permission risks and builds the permission-functionality mapping to each app, then regulates the relationship between permission and app functionality. Permizer is the first module to achieve a balance between privacy protection and app functionality under the personal privacy preference condition. Finally, we develop Permizer as a one-button service on the real-world Android OS with 58 apps. Case studies conducted on TikTok and Amazon Alexa also demonstrate its practicability and effectiveness.**

***Index Terms*—Mobile application, permission management, privacy protection, risk estimation.**

I. Introduction

smart appliances and smart cars) manufactured by different vendors, provide user interfaces, and ensure the working logic of the system.

Despite the convenience brought by mobile applications, they also introduce serious privacy leakage issues [3]. In 2019, hundreds of thousands of users suffered breaches of private data via multiple companies, including Facebook, Orvibo, and Blur [4]. Among the various reasons of privacy leakage [5], [6], the most common reason is that app developers needlessly request permissions to gather user information, the so-called *over-privileged phenomenon* [7]. Such excessive permissions requesting is a pervasive phenomenon among mobile applications, which potentially undermines user privacy. One example is Baidu Map, a popular app used in smart transportation systems. It requests 13 sensitive permissions, including location, contact information and camera access, when starting the service. Although requesting location permission is reasonable for the app's navigation function, contact information, and camera will not be used in most scenarios. Granting these less relevant permissions is unnecessary and highly risky. To solve such a security issue, an efficient and user-friendly application permission management scheme is urgently needed.

Traditional permission management schemes were



Abstract

- Single paragraph
- If not properly structured or written, could cause editor rejection.
- Summarizes the problem, methods, results, and conclusion
- Should be:
  - Interesting & understandable
  - Accurate & specific (editor might build decision based on this)
  - Concise

Fast Consensus-Based Time Synchronization Protocol Using Virtual Topology for Wireless Sensor Networks

Linh-An Phan, *Student Member, IEEE*, and Taehong Kim, *Member, IEEE*

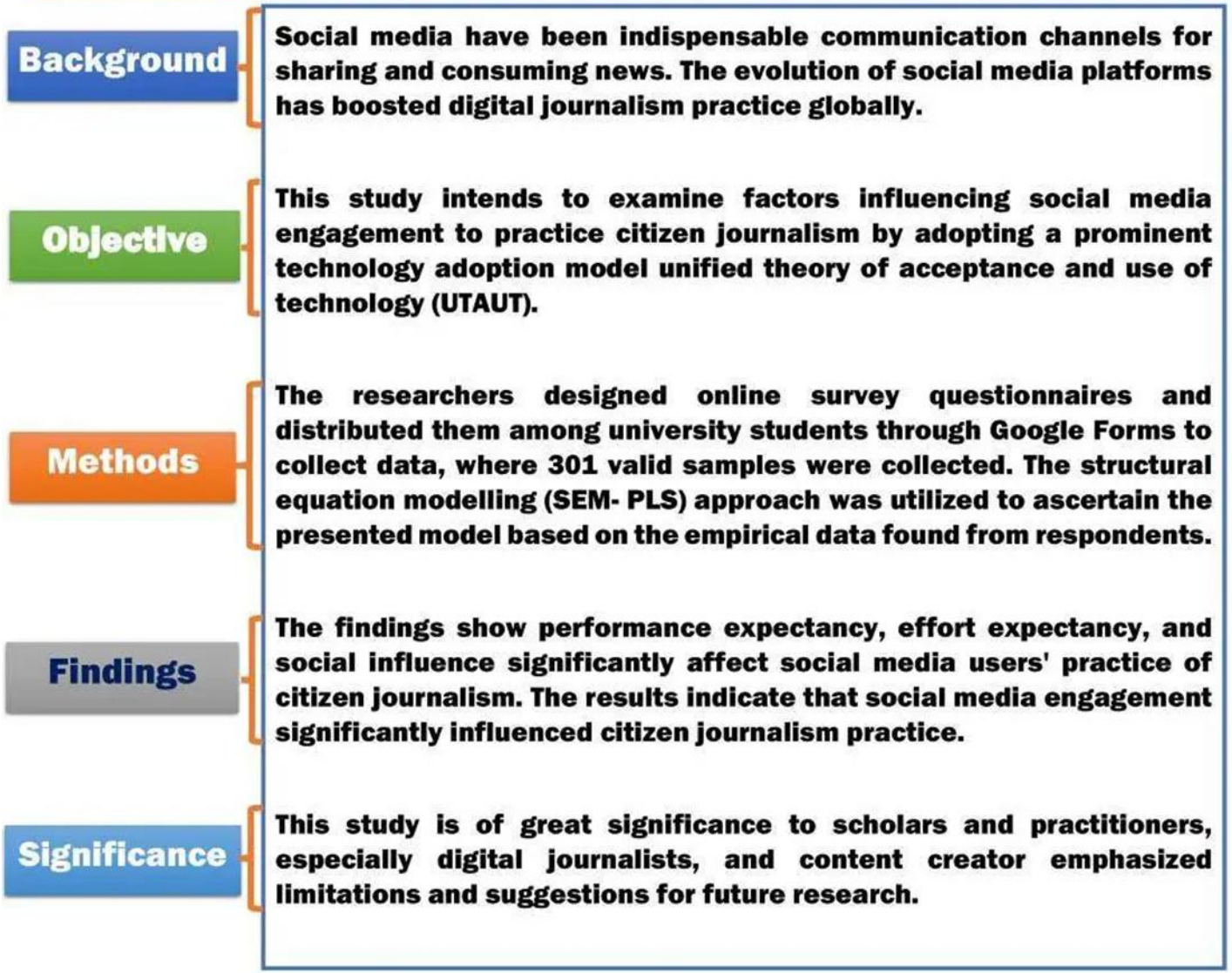
***Abstract*—Slow convergence is a major drawback of average-based consensus time synchronization protocols, particularly in large or sparse wireless sensor networks. The convergence speed can be increased by adding more nodes or increasing the transmission range of nodes, because the network becomes strongly connected. However, these solutions are not always feasible owing to hardware constraints. In this article, a virtual topology-based time synchronization protocol (VTSP) is proposed to address the aforementioned drawback of consensus-based protocols. Notably, VTSP performs the consensus process on a virtual topology that has a stronger algebraic connectivity than a physical one. Therefore, VTSP can significantly accelerate convergence speed without modifying the physical structure of the network. The virtual topology is formed by creating virtual links between each node and its two-hop neighbors. Moreover, two optimization techniques are suggested for reducing data redundancy, which is caused during the creation of the virtual links, in timing messages and for speeding up the convergence by excluding edge nodes from the consensus process. Simulation results demonstrate that VTSP can achieve convergence three times faster than the gradient time synchronization protocol, a well-known consensus-based time synchronization protocol, in various topologies while preserving the same level of accuracy.**

***Index Terms*—Algebraic connectivity, average consensus, fast convergence, time synchronization, virtual link, wireless sensor network (WSN).**

method to perform the consensus in distributed time synchronization protocols [1], [2]. However, the main drawback of average-based consensus time synchronization protocols is slow convergence (synchronization), especially in large or sparse WSNs [1], [3]. Although numerous studies regarding this type of protocol have been conducted, slow convergence is still an ongoing problem in average-based consensus time synchronization protocols.

In an average-based consensus time synchronization protocol, all nodes in the network converge to a common clock by iteratively averaging the clock values of their neighbors. This process requires many synchronization rounds to finally achieve convergence. It is important to note that the convergence speed (rate) of an average-based consensus protocol depends on the algebraic connectivity of the network [4], [5]. In particular, if a network is strongly connected, it is able to be rapidly synchronized [6]. Therefore, an obvious solution for the convergence problem of average-based consensus time synchronization protocols is to increase the algebraic connectivity of the network. For instance, more nodes must be added or the transmission range of nodes must be extended in the network. However, adding more nodes incurs additional hardware costs, while the transmis-

Research Paper Abstract Example APA





Abstract

Environmental factors determine the suitability of natural habitats for crop pests and often facilitate their proliferation and that of the crop diseases they carry. Crop pests and diseases damage food crops, significantly reducing yields for these commodities and threatening food security in developing, predominantly agricultural economies. Given its impact on environmental factors, climate change is an important determinant of crop pest and disease distribution. This study uses Targeting Tools, a climate suitability analysis and mapping toolkit, to explore the potential impact of climate change on select environmental factors linked to crop pest and associated diseases' proliferation. Based on the existing literature, prediction modeling was performed on 21 key pests and diseases that impact the major food crops for Zambian consumption. Future changes in habitat suitability for these crop pests and diseases were mapped based on their optimal temperature and relative humidity conditions for proliferation. Results project that there will be an overall increased geographical spread of suitable habitats for crop pests (and as follows, crop diseases) that thrive in warmer environments. By the 2030s, crop pests and diseases will increasingly spread across Zambia, with a higher likelihood of occurrence projected under RCPs 2.6, 4.5, and 8.5. Crop pests and diseases that thrive in cooler environments will experience decreasing habitat suitability in the 2030s, but will transition to a slower decrease in the 2050s under RCPs 2.6 and 4.5. Overall crop pest and disease habitat suitability will continue to rise slowly in the 2050s. RCP 8.5 shows an increased habitat suitability for crop pests and diseases that thrive in warm environments, with a decreased likelihood of occurrence for crop pests and diseases that thrive in cooler environments. The results highlight the need for future-facing, long-term climate adaptation and mitigation measures that create less suitable microclimates for crop pests and diseases.

Background statement

Problem statement/ rationale

Aim/objective/ purpose

Methods

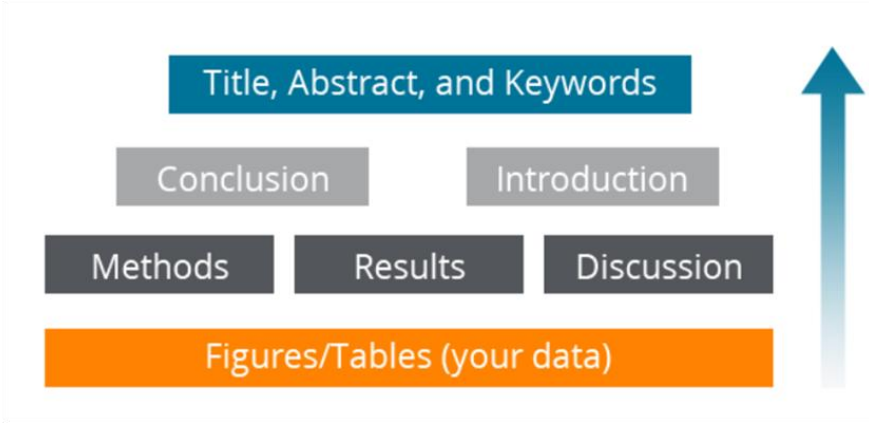
Results

Conclusions/ implications/ contribution



Main Body

- **IMRAD**
  - **I**ntroduction
  - **M**ethodology
  - **R**esults **A**nd **D**iscussion
- If any of the IMRAD elements is missing, article will get desk rejection
- When writing the article, it is **strongly** recommended to follow the bottom-up approach



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Understanding Research Paper Introductions

What Is It?

The introduction is the first section that presents your topic and sets the stage for the entire paper. It provides essential background information and clearly explains the purpose of your research while outlining the main argument or thesis statement.

Why It Matters

- Draws the reader in and encourages continued engagement
- Sets the context with necessary background information
- Explains the purpose and significance of your research
- Introduces the main idea through a clear thesis statement
- Guides the reader through what to expect

Hook

A sentence or question that grabs attention and sparks interest in your topic

Background

Brief details providing context to help readers understand the topic

Research Problem

Clearly state the issue or question your research addresses

Thesis Statement

A clear sentence outlining your main argument or focus

Purpose & Scope

Explain what your research covers and why it's important

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Key Guidelines & Best Practices



Length Recommendations

Your introduction should typically comprise about **10% of your total paper length.** For a standard 10-15 page research paper, this translates to approximately 1-1.5 pages. It should be long enough to provide context and outline objectives without overwhelming readers with excessive detail.

When to Write It

It's often best to write your introduction *after* completing the main sections like methodology, results, and discussion. This approach gives you clearer understanding of your findings, allowing you to introduce your topic more effectively. Start with a rough draft, but polish it last for optimal clarity.

Start Broad, Then Narrow

Begin with a general overview to set context, then gradually focus on your specific research issue

Define Key Terms Early

If your paper uses technical terminology, define it upfront to prevent confusion

Highlight Relevance

Explain why your research matters and how it connects to current issues or debates

Maintain Objectivity

Keep your tone academic and fact-focused, avoiding overly emotional or exaggerated claims

Stay Focused

Keep content relevant—every sentence should serve the purpose of explaining your research problem



The Introduction

Provides a brief context to the readers

Addresses the problem

Editors normally read this

Related work should be discussed

- Another approach is to put related work in another section after the introduction

Identifies the solution and limitations

Novelty should be clearly described

Editor also focus on this

Provides perspective consistent with the journal



Methods

One of the critical sections that determines whether reviewers and perhaps editor accepts or rejects the paper

Editors normally examine certain parts only

Reviewers examine it **thoroughly**

Describe how the problem was studies

Includes details information

For reproducibility

Do not describe previously published procedures

Identifies the equipment and materials used



Results

- Should be clear and easy to understand
  - Editors might read the **first paragraph** only
  - Reviewers examine it thoroughly
- Include only data for primary importance
- Highlight the main findings
- Feature unexpected findings
- Provide statistical analysis
- Include figures and illustrations

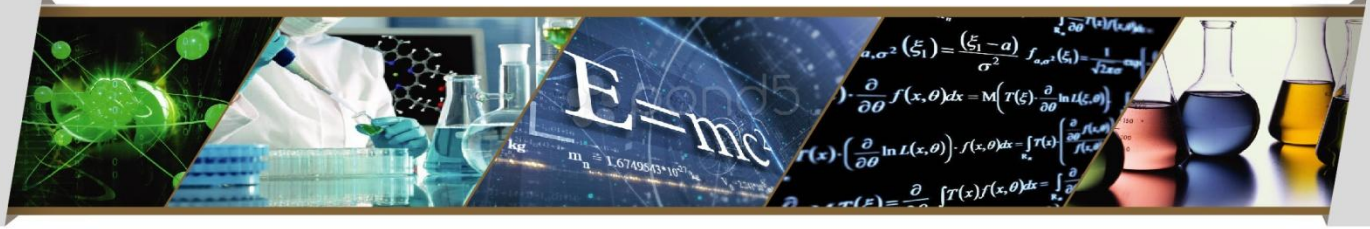
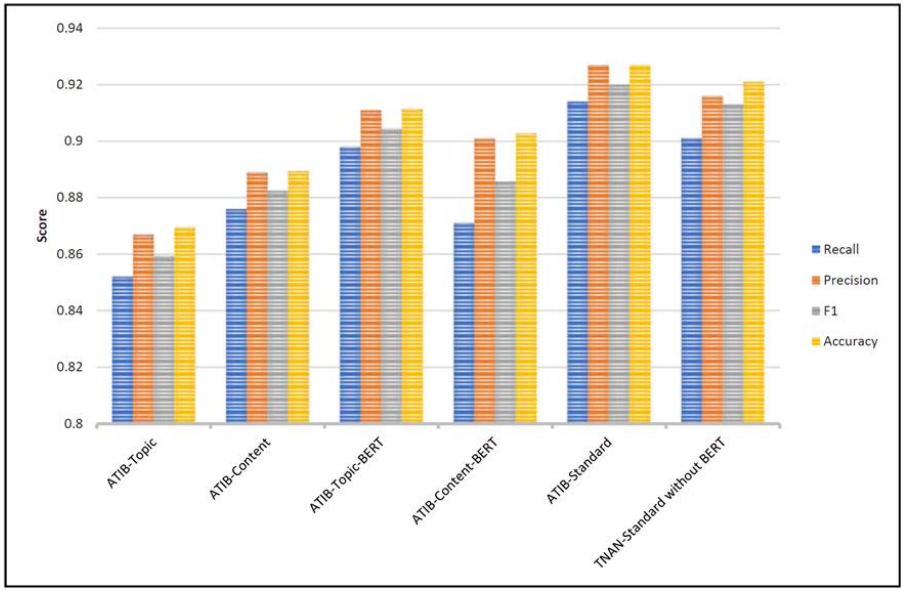
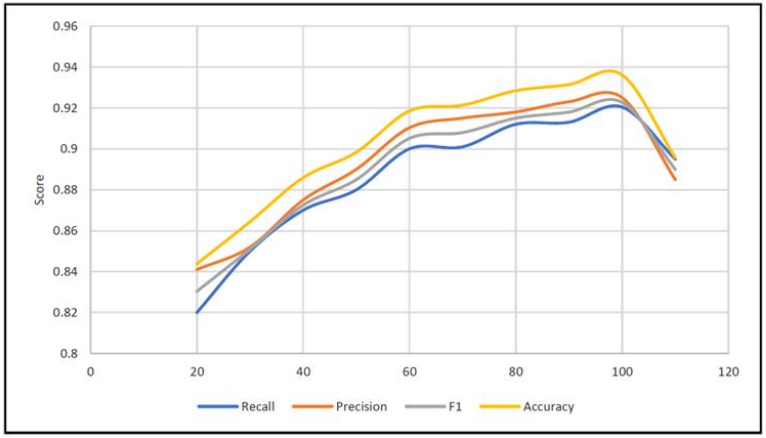


Table 3
Performance comparison of the TNAN model on Dataset 1.

Model	Rec	Pre	F1	Acc
SVM	0.672	0.821	0.739	0.748
MF [47]	0.769	0.91	0.834	0.825
TSD [3]	0.795	0.88	0.835	0.842
DLF [7]	0.826	0.871	0.848	0.855
DeepTwitter [6]	0.837	0.851	0.844	0.837
DeepSBD [8]	0.873	0.882	0.877	0.884
Bi-LSTM [32]	0.891	0.912	0.901	-
TNAN	0.914	0.927	0.920	0.927







Discussion

The most important section

Editors might read the first paragraph only

Reviewers read it thoroughly

Interprets the results

Make discussion correspond to the results and complement them

Do not repeat the results

Compare with the published results



Conclusion

Be clear

Editor and reviewers read it

Provide justification for the work

Explain how your work advances the current state of the art

Suggest for future experiments



Reference

- Do not use too many references
- Conform strictly to the style given in the Guide for Authors
  - Could lead to desk rejection or resubmission



Acknowledgment

- Thanks your team and university.
- Include research grant if used for funding or materials



Thank you

